Ex5: Information Extraction using spaCy

Aim: To extract tokens, lemmas, POS tags and named entities from customer reviews using spaCy.

Learning Objective:
The objective of this experiment is to enable students to understand and implement information extraction techniques on unstructured customer review text using spaCy.

Problem Statement:
An e-commerce company receives thousands of customer reviews containing valuable information about products, brands, organizations, locations, quantities, and other entities. Manually identifying and organizing this information from large volumes of reviews is time-consuming and inefficient.

As an NLP Engineer, your task is to process customer reviews using the spaCy Natural Language Processing library and develop an information extraction system that identifies and extracts meaningful linguistic information and named entities from the reviews. The extracted information should be organized into a structured format that can be further used for customer feedback analysis.

Tasks to Perform:

1. Reuse the Amazon Fine Food Reviews Dataset used in the previous experiment and load it into a Pandas DataFrame. Install and import the required libraries.

Clean the dataset

2. Select the Text column containing customer reviews, Remove missing or null values, Remove duplicate reviews (if any), Retain only the first 10,000 reviews for faster processing.

Text Preprocessing and Part-of-speech (POS) tags

3. Apply the spaCy pipeline to the review text. For any five reviews, display: Original review, Tokens, Lemmas, Part-of-speech (POS) tags.
4. Identify and display different POS categories such as: Nouns, Verbs, Adjectives, Adverbs.
5. Count and display the frequency of the major POS categories present in the reviews.

Named Entity Recognition (NER)

6. Apply Named Entity Recognition (NER) using spaCy to identify entities in the reviews. Extract and display entities such as: PERSON, ORG, DATE, MONEY, QUANTITY, PRODUCT, where recognized by the selected spaCy model. Display the extracted information in a structured DataFrame.

Information Extraction

7. Create a function that accepts a customer review as input and returns: Tokens, Lemmas, POS tags, Named entities, Entity labels.
8. Test the function using at least three new customer reviews.
9. Display the extracted information in a structured format.

Solution:

```
# pip install spacy pandas
# python -m spacy download en_core_web_sm
import pandas as pd
import spacy
from collections import Counter
```

```
nlp = spacy.load("en_core_web_sm")

# Load and clean the dataset
df = pd.read_csv("Reviews.csv")
df = df[["Text"]].dropna().drop_duplicates().head(10000).reset_index(drop=True)

# Tokens, lemmas and POS tags for five reviews
for text in df["Text"][:5]:
    doc = nlp(text)
    print("Review:", text)
    print("Tokens:", [t.text for t in doc])
    print("Lemmas:", [t.lemma_ for t in doc])
    print("POS   :", [(t.text, t.pos_) for t in doc], "\n")

# POS categories for the first review
doc = nlp(df["Text"][0])
for name, tag in [("Nouns", "NOUN"), ("Verbs", "VERB"),
                  ("Adjectives", "ADJ"), ("Adverbs", "ADV")]:
    print(name, ":", [t.text for t in doc if t.pos_ == tag])

# POS frequency and NER over all reviews
docs = list(nlp.pipe(df["Text"], batch_size=200, disable=["parser"]))
pos_count = Counter(t.pos_ for d in docs for t in d
                    if not t.is_punct and not t.is_space)
print(pd.DataFrame(pos_count.most_common(), columns=["POS", "Count"]))

wanted = {"PERSON", "ORG", "DATE", "MONEY", "QUANTITY", "PRODUCT"}
rows = [{"Review_ID": i, "Entity": e.text, "Label": e.label_}
        for i, d in enumerate(docs) for e in d.ents if e.label_ in wanted]
ner_df = pd.DataFrame(rows)
print(ner_df.head(20))
print(ner_df["Label"].value_counts())

# Information extraction function
def extract_info(review):
    doc = nlp(review)
    return {
        "Tokens": [t.text for t in doc],
        "Lemmas": [t.lemma_ for t in doc],
        "POS": [(t.text, t.pos_) for t in doc],
        "Entities": [e.text for e in doc.ents],
        "Entity Labels": [(e.text, e.label_) for e in doc.ents],
    }

tests = [
    "I bought Nestle coffee for $12 in January and loved the taste.",
    "The 2 lb pack from Amazon arrived late, but John said it was fresh.",
    "Kellogg's cereal costs 5 dollars at Walmart and my kids enjoy it.",
]
for t in tests:
    info = extract_info(t)
    print("\nReview:", t)
    print(pd.DataFrame({k: [v] for k, v in info.items()}).T)
```

Result:

Tokens, lemmas, POS tags and named entities were extracted from the reviews using spaCy and displayed in a structured format.